

# Generalizable PDL Regression via Transfer Learning in Optical Links

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**Abstract** We demonstrate transfer learning for PDL regression by adapting a simulation-trained model to experimental data under unseen operating conditions. Across different symbol rates, launch powers, and modulation formats, it consistently outperforms a source-only baseline, demonstrating robust cross-domain generalization with few target samples. ©2026 The Author(s)

## Introduction

Polarization-dependent loss (PDL) is a critical impairment in coherent optical systems, as it introduces amplitude imbalance between the two polarization tributaries and increases performance penalties and system margin requirements [1-5]. Our previous studies [3-5] showed that the statistical distribution of signal-to-noise ratio (SNR) carries a distinct fingerprint of PDL. In particular, SNR-based features collected at a single receiver were shown to support machine-learning-based localization and estimation of PDL-related impairments. However, these models generally need to be retrained when the link configuration, launch power, modulation format, or experimental platform changes, which limits their practical scalability.

Transfer learning has recently emerged as an effective way to adapt neural networks to new tasks or domains with limited labeled data [6,7]. This is particularly relevant to optical performance monitoring, where experimental data collection is costly and time-consuming. In this paper, we extend our prior SNR-statistics-based PDL analysis by proposing a transfer-learning framework for cross-domain PDL regression. A single simulation-trained source model is adapted to experimental data from updated setups and previously unseen operating conditions, using only a few target-domain samples.

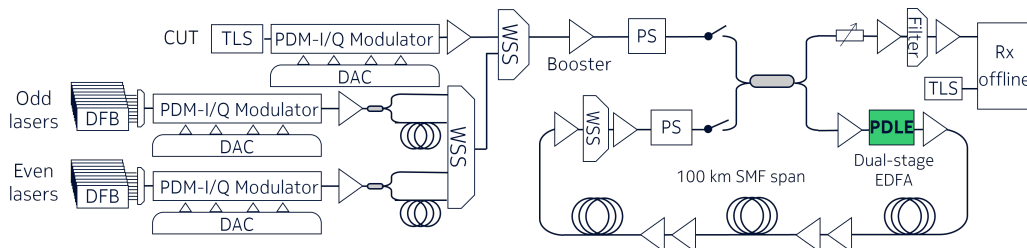
## Experimental testbed and measurements

The experimental testbed is shown in Fig. 1. On the transmitter side, a 15-channel wavelength-division multiplexing (WDM) signal with 50 GHz spacing is generated. Three modulators generate

the channel under test (CUT), even channels, and odd channels. The CUT is produced by a tunable laser source (TLS) at 1545.3 nm, while the even and odd channels are generated by two distributed feedback laser (DFB) arrays. All channels are pulse-shaped with a root-raised-cosine filter (roll-off 0.01). The even, odd, and CUT channels are combined using wavelength-selective switches (WSS) to form the final WDM signal.

The WDM signal is launched into a three-span standard single-mode fiber (SMF) link. Each span is 100 km long, with a chromatic dispersion of 16.7 ps/nm/km, an attenuation coefficient of 0.22 dB/km, and a nonlinear coefficient of 1.3 1/W/km. Polarization-mode dispersion is neglected. A polarization scrambler (PS) is placed before the PDL emulator (PDLE) to randomize the input state of polarization. The dominant PDL is then introduced by the PDLE, highlighted in green in Fig. 1, which is placed at the inter-stage of a dual-stage erbium-doped fiber amplifier (EDFA).

At the receiver (Rx), only the CUT is monitored. For each polarization tributary, 7000 SNR traces are collected. Each trace is processed using digital chromatic-dispersion compensation, polarization demultiplexing based on the constant-modulus algorithm (25 taps), and carrier-phase recovery [8]. The SNR of each tributary is then estimated from the variance of the constellation clouds. Measurements were performed for 32-, 39-, and 46-Gbaud QPSK at launch powers from -5 to -1 dBm. To assess generalizability across modulation formats, 32-Gbaud 16QAM was also tested. All operating conditions were kept within the linear regime, where nonlinear Kerr effects were negligible compared to ASE



**Fig. 1:** Experimental testbed. PDM-I/Q: polarization-division multiplexed in-phase quadrature; DAC: digital-to-analog converter.

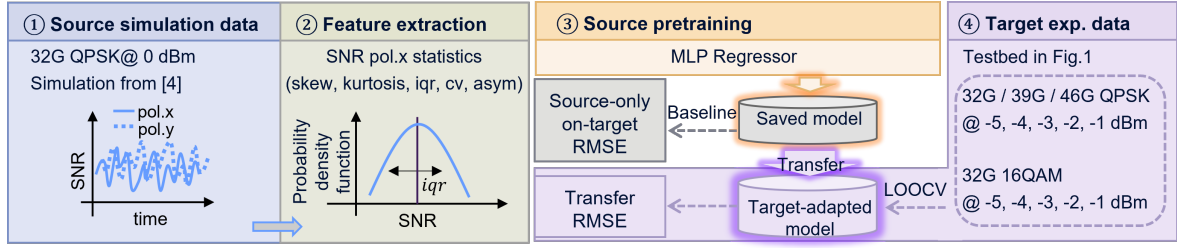


Fig. 2: Transfer learning pipeline for cross-domain PDL regression.

noise.

### Transfer learning framework

Figure 2 summarizes the proposed transfer-learning pipeline. It consists of four stages: (1) source-domain simulation data preparation, (2) statistical feature extraction from SNR distributions, (3) source-domain pretraining of a multi-layer perceptron (MLP) regressor, and (4) transfer of the pretrained model to the target experimental domain for leave-one-out cross-validation (LOOCV)-based adaptation and evaluation [9]. The goal is to reuse a single simulation-trained PDL estimator across new experimental conditions without rerunning split-step Fourier method (SSFM) simulations for every setup.

In the source domain, we use the same 32-Gbaud QPSK simulation dataset as in our previous study [4], which was generated using a SSFM-based propagation model. The EDFA gains and component losses were configured to match the experimental link budget, and EDFA noise figures are taken from datasheets. Since the exact PDL values of the WSSs and EDFAs are unavailable, each of them is assigned a fixed value of 0.1 dB in the simulation. The average power loss introduced by the PDLE was compensated by dynamically adjusting the EDFA gain. For each polarization tributary, 1000 SNR samples were collected. In addition, the transceiver penalty  $\text{SNR}_{\text{TRX}}$  was characterized in back-to-back measurements and removed from the measured SNRs so that the analysis focuses on ASE-noise-induced SNR fluctuations. This simulation dataset was generated once and then re-used as the source domain for all subsequent target-domain transfers.

As an initial proof of concept, we consider only the SNR distribution of the x-polarization tributary, from which five statistical descriptors are extracted, namely skewness, kurtosis, interquartile range (iqr), coefficient of variation (cv), and asymmetry (asym). The last two features are defined as

$$\text{cv} = \frac{\text{std}}{|\text{mean}| + \varepsilon} \quad (1)$$

$$\text{asym} = \frac{\text{max} - \text{mean}}{\text{mean} - \text{min} + \varepsilon} \quad (2)$$

where std, mean, max, and min denote the standard deviation, mean, maximum, and minimum of the SNRs, respectively, and  $\varepsilon = 10^{-8}$ . These features summarize the shape of the SNR distribution and form the input of the regression model.

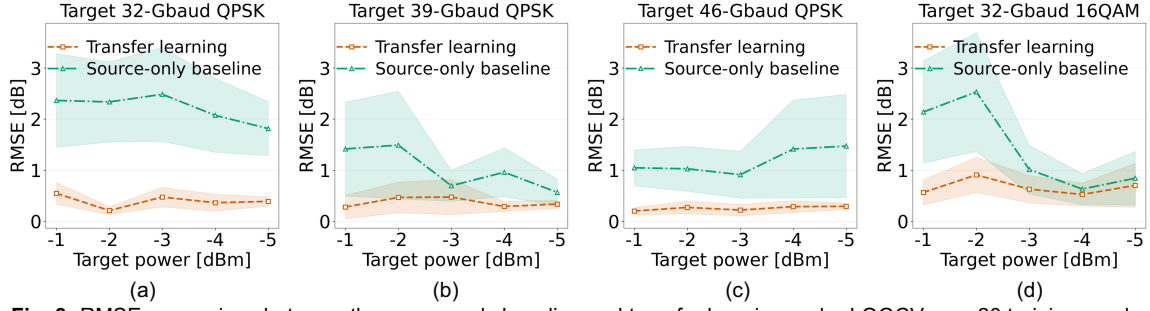
In step (3), “Source pretraining” in Fig. 2, an MLP regressor was trained on the simulation features for 300 epochs. For each random seed, the source dataset was split into 70% training and 30% validation subsets, and the validation RMSE was monitored on the source validation set. The resulting pretrained network, labeled as the “saved model” in Fig. 2, was stored as the source-domain model and used to initialize transfer learning, instead of training a new regressor from scratch.

In step (4), which introduces the “Target exp. data” in Fig. 2, experimentally measured target-domain data corresponding to Fig.1 were used for cross-domain evaluation. The saved model was first directly applied to these data without adaptation, and the resulting source-only on-target RMSE was used as the baseline for comparison. The target-domain dataset includes 32-Gbaud QPSK, 39-Gbaud QPSK, 46-Gbaud QPSK, and 32-Gbaud 16QAM, with launch powers from -5 to -1 dBm and PDL values ranging from 2 to 5 dB.

To evaluate transfer performance on the small target-domain dataset, LOOCV was performed separately for each target operating condition, defined by a fixed symbol rate and launch power. For each operating condition and each PDL value, the measured SNR samples were summarized using the five statistical descriptors defined above, forming one feature vector that was treated as one regression sample. In each LOOCV iteration, one regression sample was held out for testing, while the remaining samples from the same operating condition were used to fine-tune the pretrained source model, yielding the target-adapted model shown in Fig. 2. Thus, target-domain adaptation was carried out under a limited-sample setting, without requiring extensive experimental data collection. The transfer RMSE was then computed over the prediction errors from all LOOCV iterations.

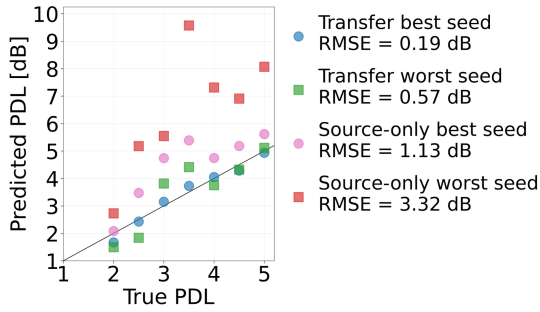
### Results and comparative analysis

To assess the sensitivity of the proposed



**Fig. 3:** RMSE comparison between the source-only baseline and transfer learning under LOOCV over 20 training seeds. Each panel corresponds to one target operating domain, and each curve shows the mean RMSE at a given launch power. Shaded regions denote one standard deviation across seeds.

transfer-learning-based PDL regression pipeline to source-domain training randomness, we repeated the full source-domain training and target-domain LOOCV evaluation over 20 different random seeds. This procedure yielded 20 pretrained source-domain models. Although seed-to-seed variations were observed, the source-domain validation error remained low across all seeds, with validation RMSE values consistently below 0.18 dB and reaching 0.03 dB in the best case.



**Fig. 4:** Predicted versus true PDL for the experimental 32-Gbaud QPSK target at -5 dBm for transfer learning and the source-only baseline under the best and worst seeds.

Figure 3 summarizes how source-domain training variability propagates to the target-domain experimental evaluation. The curves represent the mean RMSE over the same 20 training seeds used for source pretraining, while the shaded regions denote one std across the seeds. For each target domain and launch power, the source-only on-target RMSE and the transfer RMSE are compared.

Overall, transfer learning consistently outperforms the source-only baseline across all target domains and all tested launch powers in Fig. 3. The gain is particularly pronounced for the QPSK targets. For example, in Fig. 3(a), for 32-Gbaud QPSK, the mean transfer RMSE ranges from 0.21 to 0.55 dB, whereas the source-only on-target RMSE ranges from about 1.81 to 2.49 dB, corresponding to a more than fourfold RMSE reduction.

As a representative example, Fig. 4 zooms into the -5 dBm operating point in Fig. 3(a), showing the predicted versus true PDL values for

transfer learning under the best and worst seeds, together with the corresponding source-only baselines. The transfer RMSEs are 0.19 dB and 0.57 dB for the best and worst seeds, respectively. Under the same two seeds, the source-only baseline yields RMSEs of 1.13 dB and 3.32 dB.

A similar trend is observed in Fig. 3(b) for 39-Gbaud QPSK, where transfer learning consistently outperforms the source-only baseline across all tested launch powers.

In Fig. 3(c), for 46-Gbaud QPSK, the mean transfer RMSE remains between 0.20 and 0.29 dB, whereas the source-only RMSE is substantially higher, ranging from 0.91 to 1.47 dB. The corresponding std ranges from 0.06 to 0.13 dB for transfer learning, compared with 0.35 to 1.01 dB for the source-only baseline.

In Fig. 3(d), the 32-Gbaud 16QAM target is more challenging and exhibits larger variability, likely due to the additional domain shift introduced by the modulation-format mismatch. Nevertheless, transfer learning still achieves a lower mean RMSE than the source-only baseline at all tested power levels, with the mean transfer RMSE remaining below 0.9 dB. The shaded regions further show that this improvement is robust across repeated source-domain training runs: the std ranges from 0.23 to 0.42 dB for transfer learning, compared with 0.30 to 1.16 dB for the source-only baseline.

## Conclusions

A transfer-learning-based PDL regression approach was experimentally validated on an optical testbed. By transferring a simulation-trained model to unseen experimental conditions, accurate PDL estimation was achieved without regenerating the source-domain dataset. Across multiple symbol rates, launch powers, and modulation formats, the proposed method consistently outperformed the source-only baseline while requiring only limited target-domain data for adaptation. These results highlight the potential of transfer learning as a practical solution for data-efficient PDL estimation in evolving optical systems.

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